WEDNESDAY - CTF WRITEUP

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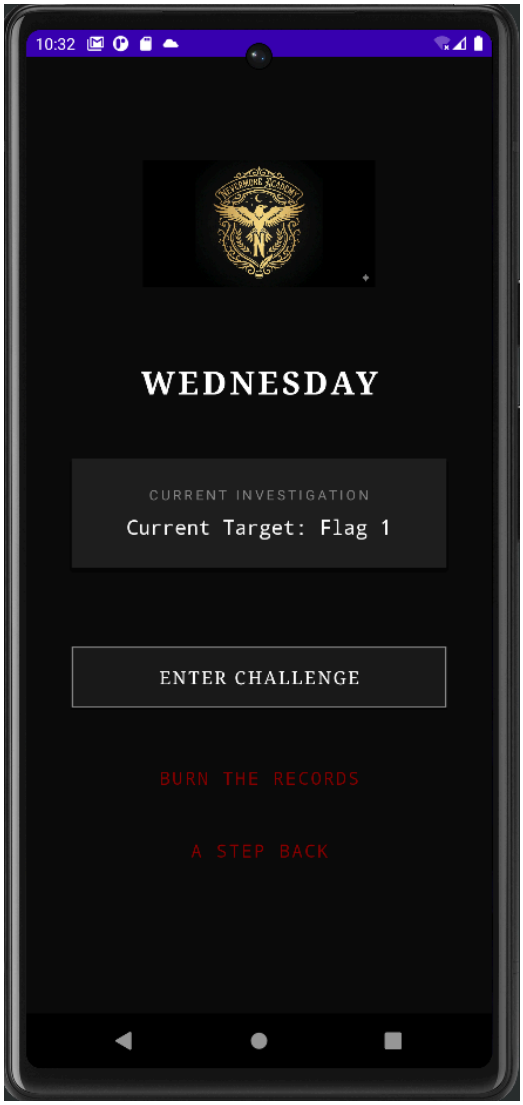
FLAG 1 - The Arrival

Strings.xml



Here, this activity TheArrivalActivity is searched

```java
/* JADX INFO: Access modifiers changed from: private */
public static final void onCreate$lambda$0(TheArrivalActivity this$0, View it) {
    Intrinsics.checkNotNullParameter(this$0, "this$0");
    this$0.checkFlag();
}

private final void checkFlag() {
    TextInputEditText textInputEditText = this.etFlagInput;
    if (textInputEditText == null) {
        Intrinsics.throwUninitializedPropertyAccessException("etFlagInput");
        textInputEditText = null;
    }
    String userInput = StringsKt.trim((CharSequence) String.valueOf(textInputEditText.getText())).toString();
    if (userInput.length() == 0) {
        Toast.makeText(this, "The void remains empty.", 0).show();
        return;
    }
    String obfuscatedData = getString(R.string.typewriter_ribbon_data);
    Intrinsics.checkNotNullExpressionValue(obfuscatedData, "getString(...)");
    byte[] decodedBytes = Base64.decode(obfuscatedData, 0);
    Intrinsics.checkNotNull(decodedBytes);
    String actualFlag = new String(decodedBytes, Charsets.UTF_8);
    if (Intrinsics.areEqual(userInput, actualFlag)) {
        Toast.makeText(this, "Correct. A new mystery awaits.", 1).show();
        ProgressManager.INSTANCE.advanceLevel(this);
        finish();
        return;
    }
    Toast.makeText(this, "Incorrect. Try again, outcast.", 0).show();
}
```

```xml
<string name="title_challenge_8">Chapter VIII: I\'m Watching You</string>
<string name="title_hub">WEDNESDAY</string>
<string name="typewriter_ribbon_data">V0VETkVTREFZe3czbGMwbTNfdDBfbjN2M3JtMHIzfQ==</string>
<string name="wv_auth_token_success">BJISJXIFD{r4hhm14y0_r0sxy3w}</string>
esources>
```

INTERPRET AS
BASE64

CONVERT TO
TEXT

```
V0VETkVTREFZe3czbGMwbTNfdDBfbjN2M3JtMHIzfQ==
```

Transform: None

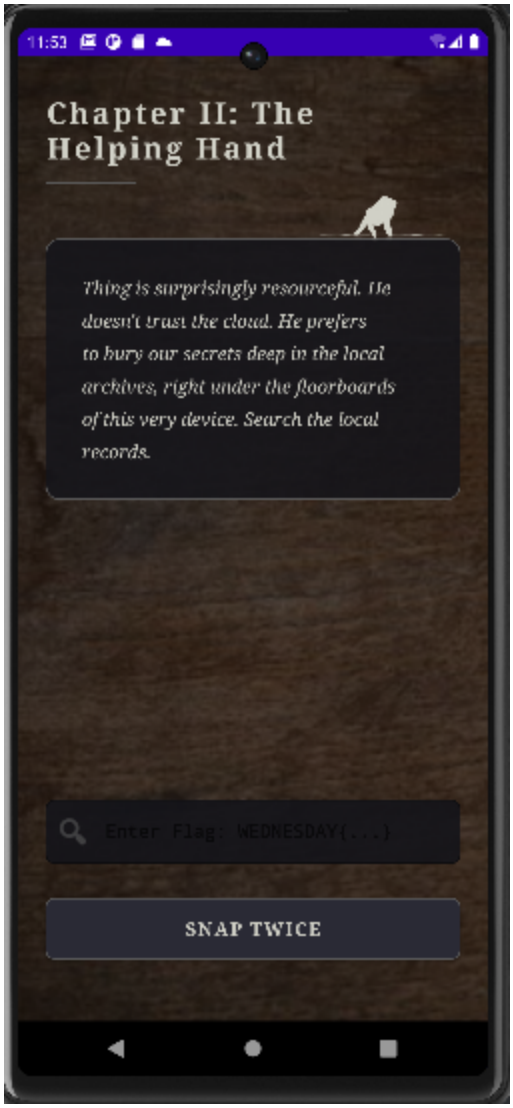
```
WEDNESDAY{w3lc0m3_t0_n3v3rm0r3}
```

When decrypted from base64 we get
WEDNESDAY{w3lc0m3_t0_n3v3rm0r3}

FLAG 2 - The Helping Hand (Thing)

We have three activities - ThingActivity, ThingHelper and CipherHelper

```java
int xorKey = cipherHelper.getKeyByName("Nightshade_Secret");
String actualFlag = decryptHexXor(encryptedHex, xorKey);
if (Intrinsics.areEqual(userInput, actualFlag)) {
    Toast.makeText(this, "Correct. Thing approves.", 1).show();
    ProgressManager.INSTANCE.advanceLevel(this);
    finish();
    return;
}
Toast.makeText(this, "Incorrect. Keep digging under the floorboards.", 0).show();
```



When the DB is analysed we get a xorkey value from Nightshade_Secret key = 42

id	key_name	key_value
1	Rowans_Prophecy	13
2	Weathervane_WiFi	99
3	Enid_Blog_Pass	7
4	Nightshade_Secret	42
5	Hyde_Trigger	104
6	Jericho_Archive	55

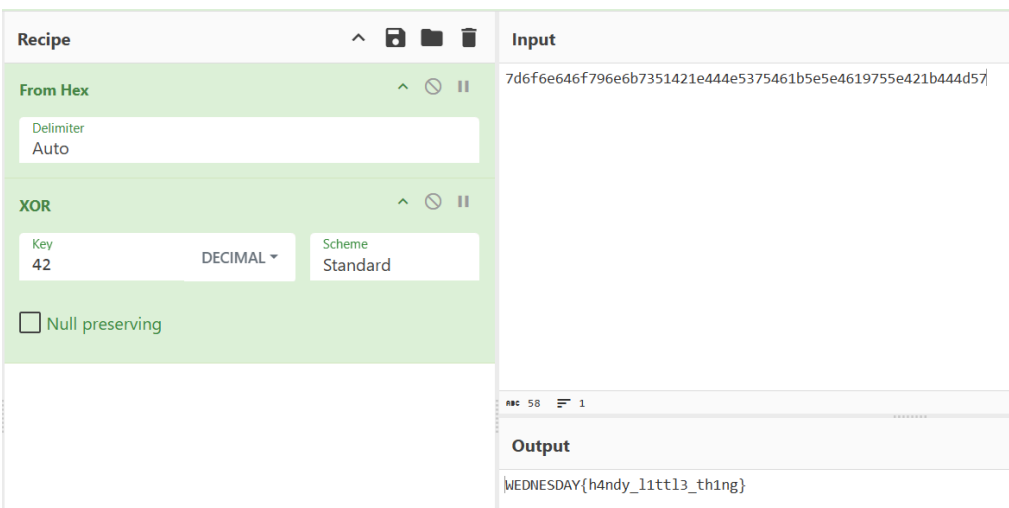
And in Cipherhelper we have a xor value list from which another key is derived

```
int[] targetBinary = {84, LocationRequestCompat.QUALITY_LOW_POWER, 105, 110, 103, 39, 39, 115, 32, 68, 105, 115, 99, 111, 118, T
Collection destination$iv$iv = new ArrayList(targetBinary.length);
for (int item$iv$iv : targetBinary) {
    destination$iv$iv.add(Character.valueOf((char) item$iv$iv));
}
String clueTarget = CollectionsKt.joinToString$default((List) destination$iv$iv, "", null, null, 0, null, null, 62, null);
Log.d("ThingHelper", "Looking for clue_name: " + clueTarget);
Cursor cursor = db.rawQuery("SELECT clue_value FROM hidden_records WHERE clue_name = ?", new String[]{clueTarget});
Intrinsics.checkNotNullExpressionValue(cursor, "rawQuery(...)");
if (cursor.moveToFirst()) {
    secret = cursor.getString(0);
    Log.d("ThingHelper", "secret : " + secret);
}
```

From the binary we get the word - Thing’’s Discovery

id	clue_name	clue_value
1	Enid''s Blog Draft	5745444e45534441597b7733723377306c665f6730737331707d
2	Rowan''s Sketchbook	5a484943485e494c5476672d695269613e527f2c6e6169522d673e70
3	Thing''s Discovery	7d6f6e646f796e6b7351421e444e5375461b5e5e4619755e421b444d57
4	Espresso Machine Logs	1026272d263027222a180053050550507c52107c0152131350157c1e
5	Tyler''s Roster	50424349425443465e7c6c343477586b37376c366960586137376b7a
6	Crackstone_Crypt_Coordinates	4e34342032332e3535312c205737322031312e323034

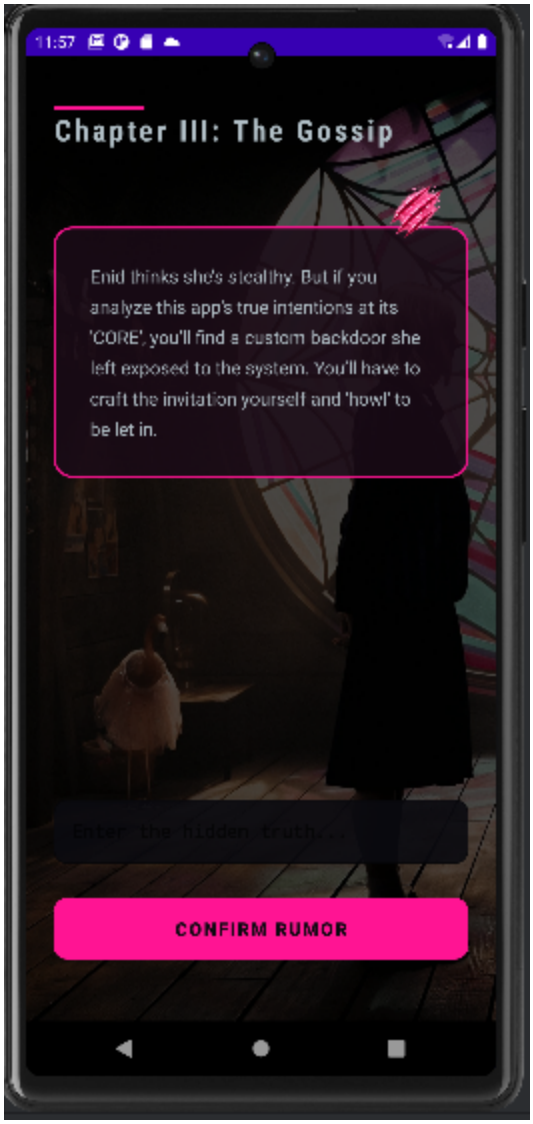
There is a hex. When we xor them we get the flag



WEDNESDAY{h4ndy_l1ttl3_th1ng}

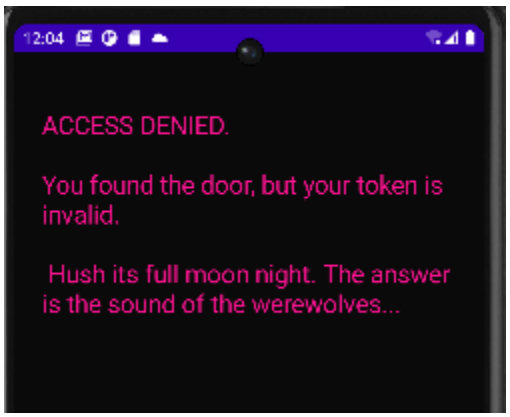
FLAG 3 - The Gossip (Enid Sinclair)

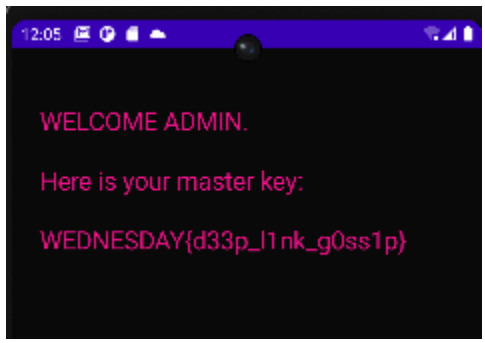
Deep links.. When looking into the clue and Enid Activity and manifest file we can look at the deeplinks given and there is the token ‘howl’ in the clue




```
<activity
    android:name="com.example.ctfwednesday.EnidAdminActivity"
    android:exported="true">
    <intent-filter>
        <action android:name="android.intent.action.VIEW"/>
        <category android:name="android.intent.category.DEFAULT"/>
        <category android:name="android.intent.category.BROWSABLE"/>
        <data
            android:scheme="nevermore"
            android:host="gossip"
            android:path="/admin"/>
    </intent-filter>
</activity>
```

```
PS C:\Users\ASUS> adb shell am start -W -a android.intent.action.VIEW -d "nevermore://gossip/admin?token=wolf"
Starting: Intent { act=android.intent.action.VIEW dat=nevermore://gossip/... }
Status: ok
LaunchState: UNKNOWN (0)
Activity: com.example.ctfwednesday/.EnidAdminActivity
WaitTime: 2880
Complete
PS C:\Users\ASUS> adb shell am start -W -a android.intent.action.VIEW -d "nevermore://gossip/admin?token=howl"
Starting: Intent { act=android.intent.action.VIEW dat=nevermore://gossip/... }
Status: ok
LaunchState: WARM
Activity: com.example.ctfwednesday/.EnidAdminActivity
TotalTime: 291
WaitTime: 310
Complete
```





WEDNESDAY{d33p_l1nk_g0ss1p}

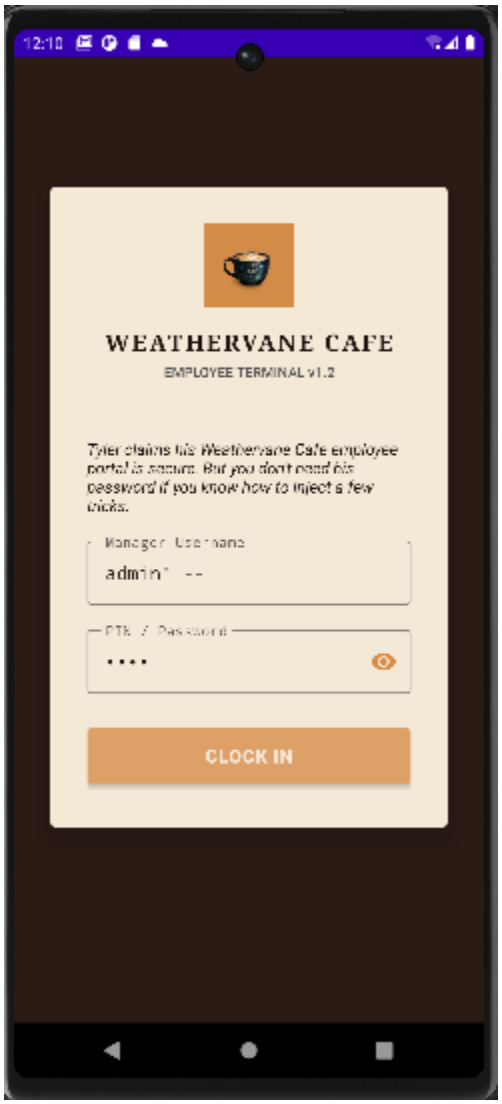
FLAG 4 - Weathervane Cafe

This is found in HydeActivity and looking at the clue we can see its a injection attack. In Weathervanedbhelper we have

```
public final String insecureLoginCheck(String username, String password) {
    Intrinsics.checkNotNullParameter(username, "username");
    Intrinsics.checkNotNullParameter(password, "password");
    SQLiteDatabase db = getReadableDatabase();
    String loggedInRole = null;
    try {
        String passwordHash = hashSHA256(password);
        String query = "SELECT role FROM employees WHERE username = '" + username + "' AND password_hash = '" + passwordHash + "'";
        Cursor cursor = db.rawQuery(query, null);
        Intrinsics.checkNotNullExpressionValue(cursor, "rawQuery(...)");
        if (cursor.moveToFirst()) {
            loggedInRole = cursor.getString(0);
        }
        cursor.close();
    } catch (Exception e) {
```

So the injection is done at the user name and the flag is appeared in the logs after the task is completed.

```
2026-04-07 12:29:14.426 28412-28412 CTF_SUCCESS com.example.ctfwednesday D ========================================
2026-04-07 12:29:14.427 28412-28412 CTF_SUCCESS com.example.ctfwednesday D SYSTEM OVERRIDE: ADMIN ACCESS GRANTED
2026-04-07 12:29:14.427 28412-28412 CTF_SUCCESS com.example.ctfwednesday D FLAG: WEDNESDAY{m4cch14t0_m0nst3r}
2026-04-07 12:29:14.427 28412-28412 CTF_SUCCESS com.example.ctfwednesday D ========================================
```

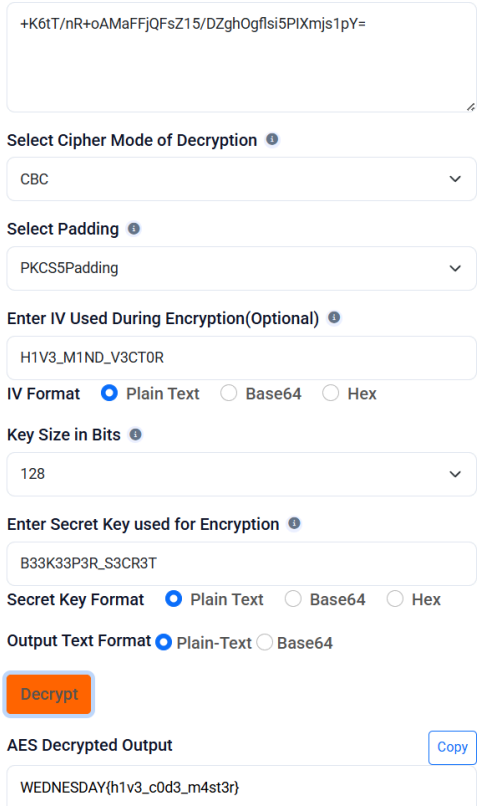


WEDNESDAY{m4cch14t0_m0nst3r}

FLAG 5 - The Hive

This is an AES based decryption flag. There is a file Hive Secret that contains the AES keys and a comment reminder to remove it

```
public final class HiveSecret {
    public static final String AES_IV = "H1V3_M1ND_V3CT0R";
    public static final String AES_KEY = "B33K33P3R_S3CR3T";
    public static final String ENCRYPTED_PAYLOAD = "+K6tT/nR+oAMaFFjQFsZ15/DZgh0gflsi5PIXmjs1pY=";
    public static final HiveSecret INSTANCE = new HiveSecret();

    private HiveSecret() {
    }
```



WEDNESDAY{h1v3_c0d3_m4st3r}

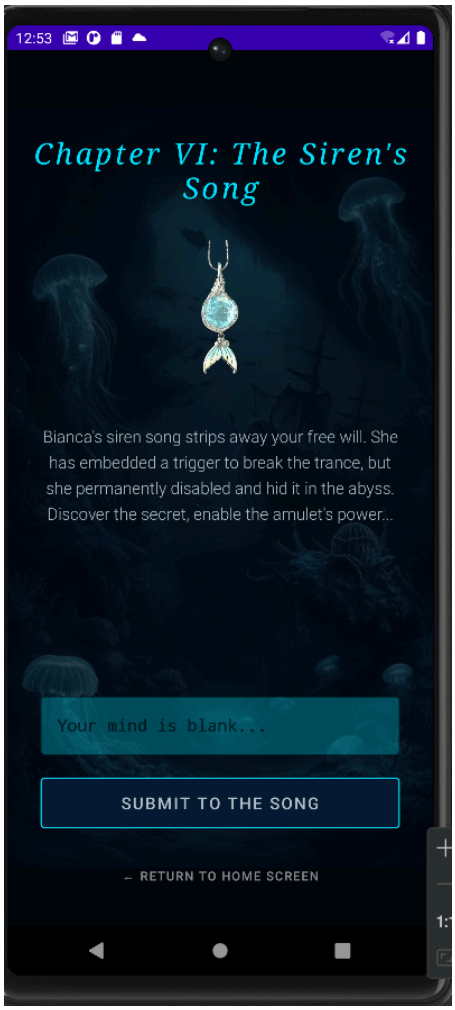
FLAG 6 - The Siren Song

This is hidden button flag which is bypassed through frida code

```
public static final void onCreate$lambda$0(SirenActivity this$0, MaterialButton $btnBreakTrance, TextInputEd
    Intrinsics.checkNotNullParameter(this$0, "this$0");
    try {
        InputStream inputStream = this$0.getResources().openRawResource(R.raw.trance);
        Intrinsics.checkNotNullExpressionValue(inputStream, "openRawResource(...)");
        byte[] encryptedBytes = ByteStreamsKt.readBytes(inputStream);
        String base64Data = StringsKt.trim((CharSequence) new String(encryptedBytes, Charsets.UTF_8)).toStri
        int yPosition = (int) $btnBreakTrance.getTranslationY();
        String dynamicKey = StringsKt.padEnd("SIREN_POS_" + yPosition, 16, '0');
        byte[] bytes = dynamicKey.getBytes(Charsets.UTF_8);
        Intrinsics.checkNotNullExpressionValue(bytes, "getBytes(...)");
        SecretKeySpec secretKeySpec = new SecretKeySpec(bytes, "AES");
        Cipher cipher = Cipher.getInstance("AES/ECB/PKCS5Padding");
        cipher.init(2, secretKeySpec);
        byte[] decodedBase64 = Base64.decode(base64Data, 0);
        byte[] bArrDoFinal = cipher.doFinal(decodedBase64);
        Intrinsics.checkNotNullExpressionValue(bArrDoFinal, "doFinal(...)");
```

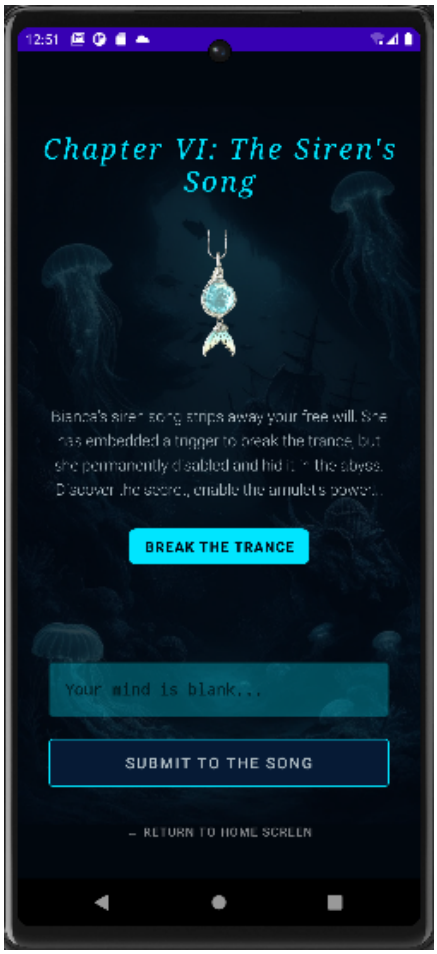
Here the AES encryption of the key is done through position of the button btnbreaktrance when set to right position 0.0.

```
Java.perform(function () {
    console.log("[*] Frida script injected successfully!");
    var View = Java.use("android.view.View");

    View.setTranslationY.implementation = function (y) {

        if (y === 5000.0) {
            console.log("[*] Banishment intercepted! A view tried to move to Y=5000.");
            console.log("[*] Forcing it back to Y=0.0...");
            this.setTranslationY(0.0);
        } else {
            this.setTranslationY(y);
        }
    };
});
```



This frida code is launched
WEDNESDAY{s1r3n_s0ng_dr34m}

```
PS C:\Users\ASUS\OneDrive\Documents\mobsec> frida -U -f com.example.ctfwednesday -l bypass.js
     ____
    / _  |   Frida 17.8.0 - A world-class dynamic instrumentation toolkit
   | (_| |
    > _  |   Commands:
   /_/ |_|       help      -> Displays the help system
   . . . .       object?   -> Display information about 'object'
   . . . .       exit/quit -> Exit
   . . . .
   . . . .   More info at https://frida.re/docs/home/
   . . . .
   . . . .   Connected to Android Emulator 5556 (id=emulator-5556)
Spawned `com.example.ctfwednesday`. Resuming main thread!
[Android Emulator 5556::com.example.ctfwednesday ]-> [*] Frida script injected successfully!
[*] Banishment intercepted! A view tried to move to Y=5000.
[*] Forcing it back to Y=0.0...
[Android Emulator 5556::com.example.ctfwednesday ]->
[Android Emulator 5556::com.example.ctfwednesday ]-> exit
```




```
byte[] bArrDecode = Base64.decode("V0VETkVTREFZe3MxcjNuX3MwbmdfZHIzNG19=", 0);
Intrinsics.checkNotNullExpressionValue(bArrDecode, "decode(...)");
String expectedHash = this$0.hashSHA256(new String(bArrDecode, Charsets.UTF_8));
Log.d("CTF_DEBUG", "Generated Hash: " + this$0.hashSHA256(userInput));
if (Intrinsics.areEqual(this$0.hashSHA256(userInput), expectedHash)) {
```

FLAG 7 - The Crackstone Crypt

This activity is for the weak RSA keys. There is a decryptflag function in the Crackstone activity

```
private final String decryptFlag() {
    try {
        String publicKeyPEM = StringsKt.replace$default(StringsKt.replace$default(StringsKt.replace$default("-----BEGIN PUBLIC KEY-----\nMIGfMA0GCSqGSI
        KeyFactory keyFactory = KeyFactory.getInstance("RSA");
        X509EncodedKeySpec keySpec = new X509EncodedKeySpec(Base64.decode(publicKeyPEM, 0));
        PublicKey publicKey = keyFactory.generatePublic(keySpec);
        Cipher cipher = Cipher.getInstance("RSA/ECB/OAEPWithSHA-256AndMGF1Padding");
        cipher.init(2, publicKey);
        byte[] decryptedBytes = cipher.doFinal(Base64.decode("qe3dQaWHa7qYzPEGkZLHFUVeH4X4ssgPx3HTS6UbBZoYO3mIHDtvHBPkxkep5kSBHYeDvZZUGKVUjt5yOm0kjYkYh
        Intrinsics.checkNotNull(decryptedBytes);
        return new String(decryptedBytes, Charsets.UTF_8);
    } catch (Exception e) {
        return "The secret remains hidden...";
    }
}
```

We use python to get the message

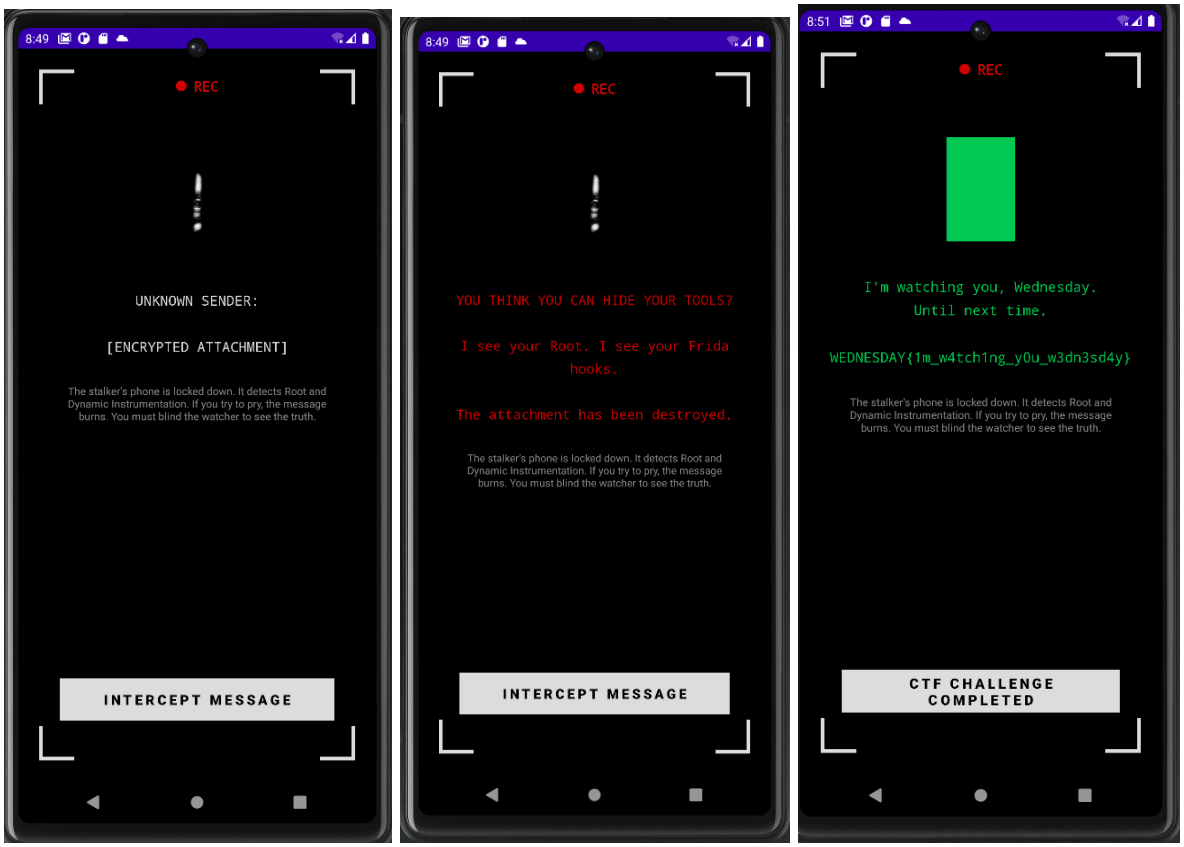
```
badcrypt.py > ...
import base64
from math import isqrt
from Crypto.PublicKey import RSA

public_pem = """-----BEGIN PUBLIC KEY-----MIGfMA0GCSqGSIb3DQEBAQUAA4GNADCBiQKBgQDAKFIWhYHFZf4+0uS+g0nor0fs+/FJeNV+pNLAPEd6dWCwx8OQJZ2nPQximgK9A0vnREdyCTB/5wP0/1wIedj0+Rqs
2mMSDtpZkvIRSINaA0OApaPNyB4TM6wQ1whLxnOslXjWt9qrsCOd3PGSfHVd4f5d5LUD/hIer0Tnd+7YYQIDAQAB
-----END PUBLIC KEY-----"""
encryptedFlag = "qe3dQaWHa7qYzPEGkZLHFUVeH4X4ssgPx3HTS6UbBZoYO3mIHDtvHBPkxkep5kSBHYeDvZZUGKVUjt5yOm0kjYkYhV2P51Cw8vUAgviJhUN3sYZlpf+gVn9+ccl3iAP+upGP9gbqUGmZ6t9oBzti5BLCX

key = RSA.import_key(public_pem)
n = key.n
e = key.e

def fermat_factor(n):
    a = isqrt(n) + 1
    while True:
        b2 = a*a - n
        b = isqrt(b2)
        if b*b == b2:
            return a - b, a + b
        a += 1

p, q = fermat_factor(n)
ciphertext_bytes = base64.b64decode(encryptedFlag)
c = int.from_bytes(ciphertext_bytes, 'big')
# 2. Calculate private exponent d
phi = (p - 1) * (q - 1)
d = pow(e, -1, phi)          # This is the modular inverse
# 3. Decrypt: m = c^d mod n
m = pow(c, d, n)

# 4. Convert back to bytes (the flag)
flag = m.to_bytes((m.bit_length() + 7) // 8, 'big')

print("Decrypted flag:", flag)
print("Flag as text:", flag.decode('utf-8', errors='ignore'))
```

This code will reveal the key of the flag
Decrypted flag: b'f1r3_w1ll_r41n_d0wn'
Flag as text: f1r3_w1ll_r41n_d0wn

WEDNESDAY{f1r3_w1ll_r41n_d0wn} is the flag

FLAG 8 - The stalker

This challenge scans the root, frida servers in the emulators machine. So we must use frida code to bypass the challenge



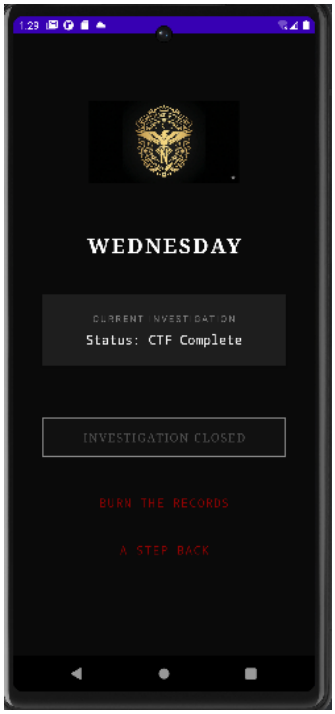
The frida code

bypass.js | badcrypt.py | scanner.js

mobsec > scanner.js > Java.perform() callback

```js
Java.perform(function () {
    var StalkerActivity = Java.use("com.example.ctfwednesday.StalkerActivity");

    StalkerActivity.isDeviceRooted.implementation = function () {
        console.log("[+] Bypassed: isDeviceRooted()");
        return false;
    };

    StalkerActivity.isFridaInjected.implementation = function () {
        console.log("[+] Bypassed: isFridaInjected()");
        return false;
    };

    StalkerActivity.isDebugEnvironmentActive.implementation = function () {
        console.log("[+] Bypassed: isDebugEnvironmentActive()");
        return false;
    };

    console.log("[*] All stalker alarms are now blind. Ready to intercept.");
});
```

The challenges are hereby completed.


1:29
WEDNESDAY
CURRENT INVESTIGATION
Status: CTF Complete
INVESTIGATION CLOSED
BURN THE RECORDS
A STEP BACK